

KEY LAKE, SK, Canada

WMO: 710330

Lat: 57.256N

Lon: 105.618W

Elev: 1686

StdP: 13.82

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.2	-38.1	-49.0	0.3	-43.4	-45.4	0.4	-37.4	19.0	4.4	16.9	2.0	1.6	270	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	80.8	61.1	77.5	59.5	74.5	58.3	63.8	75.4	62.1	72.6	60.4	70.4	9.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	82.8	66.2	58.3	77.4	64.6	56.5	72.5	63.5	30.0	75.9	28.7	72.4	27.5	70.3	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.7	14.9	-51.7	86.7	4.4	3.3	-54.9	89.1	-57.4	91.0	-59.9	92.8	-63.1	95.2		
(5)			-50.5	66.8	3.9	1.8	-53.4	68.2	-55.7	69.3	-57.9	70.3	-60.8	71.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	-7.6	-3.7	7.9	26.9	42.7	55.6	61.0	57.5	46.7	32.3	11.9	-2.7
	DBStd	27.35	16.67	15.50	16.17	13.63	9.95	7.31	5.70	6.69	8.26	9.08	13.38	16.82
	HDD50	9091	1786	1505	1305	697	268	30	1	13	158	552	1142	1634
	HDD65	13727	2251	1925	1770	1144	694	292	149	246	550	1015	1592	2099
	CDD50	890	0	0	0	3	40	197	343	246	59	2	0	0
	CDD65	48	0	0	0	0	1	8	25	13	1	0	0	0
	CDH74	760	0	0	0	0	38	155	360	179	28	0	0	0
	CDH80	121	0	0	0	0	7	24	56	29	5	0	0	0
Wind	WSAvg	6.8	5.9	6.3	6.8	7.2	7.6	7.4	6.9	7.0	7.5	7.3	6.5	5.7
Precipitation	PrecAvg	19.4	1.0	0.9	1.1	0.8	1.3	2.2	3.6	2.9	2.4	1.4	1.2	0.7
	PrecMax	23.9	2.6	3.4	2.3	1.7	3.0	5.2	6.5	5.0	5.3	2.8	2.7	1.8
	PrecMin	15.9	0.6	0.2	0.2	0.0	0.3	0.7	1.1	0.6	0.7	0.6	0.4	0.0
	PrecStd	2.1	0.4	0.6	0.6	0.4	0.7	1.1	1.3	1.0	1.1	0.6	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	35.6	47.3	65.3	78.8	82.7	85.3	83.6	77.7	61.8	41.9	32.6
		MCWB	32.2	31.9	37.4	49.0	55.6	60.8	63.3	63.7	59.5	50.9	36.8	30.2
	2%	DB	27.6	30.0	41.4	57.3	71.7	78.2	81.1	78.9	70.0	54.0	36.6	28.4
		MCWB	26.4	27.5	34.1	44.1	52.6	58.6	61.7	61.5	56.5	45.3	33.3	26.8
	5%	DB	22.5	25.2	36.4	51.4	66.2	74.5	78.1	74.9	64.7	49.1	32.4	24.3
		MCWB	21.5	23.6	31.5	40.9	50.3	56.5	60.5	59.8	53.8	43.6	30.7	23.0
	10%	DB	16.5	20.0	31.8	46.3	61.5	71.0	74.9	71.2	60.1	44.3	29.6	19.9
		MCWB	15.7	18.8	28.5	37.9	48.0	55.3	59.1	58.4	52.0	40.3	28.1	19.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.7	32.8	38.1	49.0	57.5	64.2	66.9	66.4	61.1	51.9	37.3	30.7
		MCDB	34.0	34.9	46.1	64.2	74.2	75.9	76.4	78.7	73.9	60.4	41.3	32.2
	2%	WB	26.6	27.6	34.6	44.5	54.3	61.5	64.3	63.4	57.8	47.1	33.7	26.8
		MCDB	27.5	30.2	40.4	55.7	68.0	72.2	76.1	74.5	67.6	52.2	36.3	28.2
	5%	WB	21.8	22.7	31.9	41.4	51.6	59.4	62.6	61.5	55.1	43.6	30.8	23.2
		MCDB	22.8	24.3	36.0	50.6	64.0	69.7	73.2	71.4	62.6	48.4	32.5	24.5
	10%	WB	16.1	18.1	28.5	38.4	49.0	57.3	61.0	59.7	53.0	40.6	28.1	19.2
		MCDB	16.7	19.3	31.9	45.4	59.5	67.3	71.1	68.3	58.9	44.2	29.6	19.9
Mean Daily Temperature Range	5% DB	MDBR	19.8	24.1	27.0	24.0	23.5	22.4	21.6	19.6	17.3	12.9	14.8	17.8
		MCDBR	20.6	23.7	26.4	27.4	32.0	28.7	28.0	26.6	25.2	20.5	14.8	17.0
		MCWBR	19.6	22.7	20.0	16.5	16.3	12.7	12.0	12.6	13.6	13.4	12.4	16.1
	5% WB	MCDBR	20.5	24.7	24.6	25.7	27.3	22.7	23.0	22.6	21.8	19.3	13.7	17.0
		MCWBR	19.5	22.4	19.1	15.7	15.0	11.2	10.8	11.6	12.5	12.9	11.8	16.2
Clear-Sky Solar Irradiance	taub	0.221	0.256	0.257	0.314	0.322	0.358	0.382	0.349	0.300	0.279	0.238	0.202	
	taud	2.230	2.241	2.315	2.295	2.486	2.409	2.367	2.459	2.613	2.573	2.321	2.173	
	Ebn at Noon	220	256	291	286	291	280	269	272	275	252	211	193	
	Edh at Noon	15	23	29	35	31	34	35	30	22	17	14	11	
All-Sky Solar Radiation	RadAvg	219	548	1080	1584	1795	1822	1680	1356	895	448	187	130	
	RadStd	19	32	51	102	183	156	94	118	115	65	21	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.06	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.54	-0.77	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page